



EE 307: Theory of Measurement

Course Code: EE 307

Course Name: Theory of Measurement

L-T-P-C: 3-0-0-3

Prerequisites: Basic electrical circuits, Understanding of basic probability and statistics.

Students intended for: B.Tech.

Elective or Compulsory:

Approval: 5th Senate

Course Contents

- **Module 1: Characterization of Measurement Systems:** Significance of Measurements, Units and Standards, Instruments and Measurements, Examples of Measurement System, Instrument Characteristics (Accuracy, Range, Linearity, Sensitivity, Calibration), Dynamic Characteristics (Zero, First and Second Order Systems).
- **Module 2: Error and Noise Analysis:** Errors, Classification of Errors, Review of Probability and Statistics, Statistical Treatment of Data, Regression Analysis (Least Squares), Uncertainty Analysis.
- **Module 3: Primary Sensing Elements:** Strain Gauges, Load Cells, Linear Variable Displacement Transformers (LVDT), Potentiometers, Capacitive Transducers, Thermistors, Thermocouple, Pressure Sensors, Flow Sensors, Piezoelectric transducers.
- **Module 4: Signal Conditioning Circuits:** Analog Signal Conditioning: Amplifiers, Voltage Comparator and Filters. Sampling, Quantization, A/D Converter, D/A Converter, Digital Voltmeter, Data Acquisition, Smart Sensors.

Text Books:

1. Ernest Doebelin, **Measurement System: Application and Design**, 5th Edition, McGraw.
2. Richard S. Figliola and Donald E. Beasley, **Theory and Design for Mechanical Measurements**, Wiley & Sons.
3. D. Patranabis, **Principal of Industrial Instrumentation**, 3rd Edition, McGraw-Hill.